Stroke

TOPICAL REVIEW

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Update on Rehabilitation After Stroke: Global Changes and the Continued Importance of Therapy Intensity, Dose, and Timing

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ABSTRACT: Stroke rehabilitation research has seen an unprecedented expansion in the volume of randomized controlled trials. This sets the stage for conducting meta-analyses and network meta-analyses, which help compensate for the large number of smaller phase 2 studies. Over half of randomized controlled trials are conducted in the chronic phase when neuroplasticity is reduced, and the timing of the intervention is not directly generalizable to clinical care. Meanwhile, research productivity has plateaued in high-income countries over the past decade. Therapy intensity and dosage are regarded as important to improving recovery, but the optimal intensity and dose are still uncertain. However, timing also matters when it comes to therapy intensity and dosage. Therapy conducted in the chronic phase, when neuroplasticity has declined, requires a much higher dose to have a comparable impact as the same therapy delivered in the acute/subacute phase. More randomized controlled trials are now being conducted in low- and middle-income countries, focusing on different outcome measures and more acute/subacute randomized controlled trials when compared with trials from high-income countries.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: activities of daily living ■ developing countries ■ neuronal plasticity ■ rehabilitation research ■ stroke

Stroke rehabilitation research continues to move forward supported by an unprecedented number of randomized controlled trials (RCTs). It is estimated that there are ≈4000 RCTs of stroke motor-cognitive rehabilitation interventions in peer-reviewed English-language journals and that number is growing. Stroke rehabilitation research is complicated by its multidisciplinary nature and a plethora of different interventions and outcome measures studied against a backdrop of changing neuroplasticity over time. Stroke rehabilitation RCTs are further challenged by small sample sizes and the heterogeneity of interventions and outcome measures. Most RCTs are phase 2 trials, and relative to the high number of RCTs, stroke rehabilitation lacks Phase 3 trials. The low number of phase 3 trials is related to the need for therapy training to consistently adhere to therapy protocols, tight inclusion criteria to ensure a homogeneous study group, standardized care in control groups, the need for very large sample sizes to detect a significant change with current outcome measures, and the cost to conduct large trials. This occurs against a changing global research environment,[1] discussed later in the article.

There has been recognition that current therapy intensity and dose of rehabilitation therapies in most clinical settings are far too low, and how the enhanced neuroplasticity window seen in the acute/subacute phase of stroke leads to the largest responses to most therapeutic interventions. Over half of motor and cognitive stroke rehabilitation RCTs have been conducted in the chronic phase, where in most cases, higher doses of intervention are required to achieve a similar therapeutic effect as interventions conducted in the acute/subacute phase. The present review presents an update on poststroke

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rehabilitation using data from a comprehensive database of motor and cognitive RCTs developed by conducting and combining 5 systematic reviews up to the end of December 2024.

EVIDENCE FOR STROKE REHABILITATION IS INCREASING

Stroke rehabilitation is a program of therapies and activities, typically delivered by a multidisciplinary team including therapists, designed to reverse or compensate for stroke-related deficits and minimize subsequent complications, with the goal of maximizing functional independence. Stroke rehabilitation research evaluates how therapies, activities, and technologies can better reverse or compensate for these deficits and complications.

Our database includes 3602 RCTs published in English-language peer-reviewed journals, 3061 motor rehabilitation RCTs, and 596 cognition RCTs (Table S1) updated to the end of 2024 (Figure 1). Motor and cognitive RCTs together represent over 3 quarters of all stroke rehabilitation RCTs and deal with those deficits (motor paresis, dexterity and balance disorders, aphasia, neglect, and general cognition) which are significantly impacted by therapeutic interventions and intervening neuroplasticity. Sample sizes tend to be small (mean 51.4; median 34 participants), with a total of 185 156 participants across 3602 RCTs. Of these, only 16 RCTs (0.4%) had at least 500 participants, and 339 RCTs (9.4%) had at least 100 participants. The majority of RCTs were single-site trials, with only 477 (13.2%) RCTs conducted at multiple sites. These motor and cognitive RCTs employed 859 different outcome measures. Almost a third of studies originate from low- and middle-income countries (LMICs), and more than half were conducted in the chronic phase (≥6 months).

Interventions

The evidence on stroke rehabilitation interventions comes in the form of over 3600 motor and cognitive RCTs. As an example, from 1970 to the end of 2024, a subset of 1623 RCTs in our database examined upper extremity stroke rehabilitation motor interventions. Table 1 shows interventions utilized in >50 upper extremity motor RCTs.

Similar data sets, within our database, have been created for lower extremity[2] (1548 RCTs) and cognitive rehabilitation[3] (596 RCTs). The extensive evidence base for stroke rehabilitation makes it feasible to conduct network meta-analyses based on some of the more common outcome measures; an example is shown below, utilizing the very commonly used Fugl-Meyer Assessment-Upper Extremity (FMA-UE; Figure 2).[4] Adjunct therapies in motor RCTs, defined as “treatments that aim to modify the effect of primary behavioral intervention,” are often not utilized in clinical stroke rehabilitation despite an extensive database.[5]

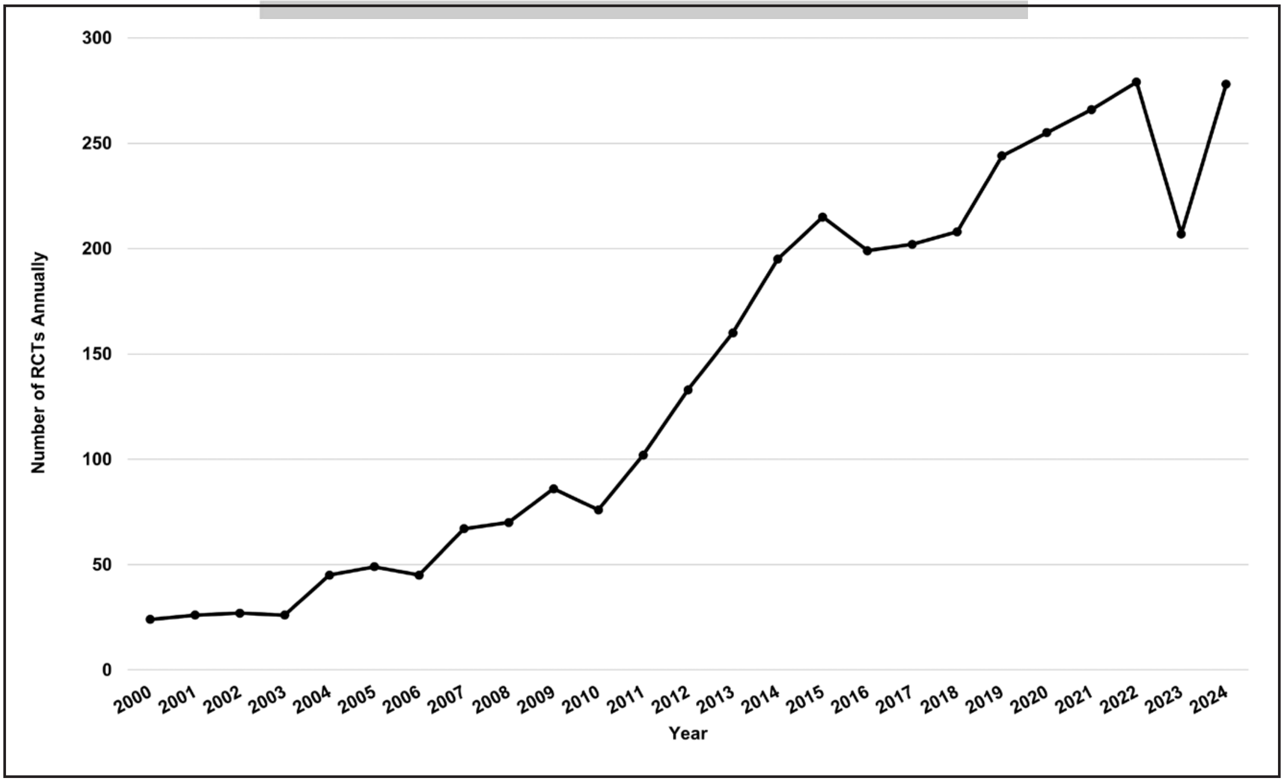

Figure 1. Stroke rehabilitation motor-cognitive randomized controlled trials (RCTs) per annum 2000–2024 (English publications) with exponential growth apart from the COVID dip in 2023.

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Table 1. Interventions Studied in Upper Extremity Motor RCTs up to the End of 2024

Interventions	No. of RCTs	Mean sample size
Robotics	238	42.4
Task-specific training	173	41.4
Virtual reality training	154	40.6
Exercise/custom physiotherapy	124	59.6
Repetitive transcranial magnetic stimulation	120	44.2
Constraint-induced movement therapy	120	37.3
Transcranial direct current stimulation	119	32.5
Mirror therapy	94	39.7
Botulinum toxin	77	72.6
Functional electrical stimulation	75	29.8
Neuromuscular electrical stimulation	72	44.6
Mental practice/motor imagery	67	28.8
Bilateral arm training	56	36.9
Orthotics/splint/assistive devices	54	35.5
Acupuncture	51	96.1
Total	1623	46.2

RCT indicates randomized controlled trial.
*Includes modified Constraint-induced Movement Therapy.

However, the problem is not a lack of research. Two examples, noninvasive brain stimulation and robotics, both regarded as adjunct therapies, have been examined in depth, studied in 510 and 391 motor and cognitive stroke rehabilitation RCTs, respectively (Tables 2 and 3) with only sporadic clinical use.

Noninvasive Brain Stimulation

Most trials today in noninvasive brain stimulation have focused on variations using 2 different stimulation modalities, repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDCS). There are several different variations in the administration of these modalities that have the potential to impact outcomes, which has, in part, contributed to the substantive number of publications in the area. The frequency of stimulation in rTMS and the location of the anode/cathode in tDCS are thought to influence the excitatory or inhibitory nature of the stimulation on the brain. The impairment being targeted (eg, upper extremity motor function), and the specific stimulation protocol influence where/on which side of the brain the researchers target stimulation. In many cases, brain stimulation is paired with another therapy (eg, exercise), either immediately after stimulation (rTMS) or during stimulation (tDCS), with the underlying hypothesis that the stimulation should augment the effects of exercise therapy. In general, rTMS is a more focused stimulation method, able to target smaller structures in the brain such as M1 (motor cortex), whereas tDCS is typically less focal with significant current spread across larger portions of the brain. Methods to provide more focused high-definition tDCS are being tested as a potential means of improving efficacy.[6] Newer and more focal techniques allowing access to deeper brain structures have emerged (eg, focused ultrasound and transcranial interference stimulation); however, few trials in stroke are available currently.[7]

Evidence for the use of rTMS for poststroke motor impairment has grown significantly over the past several years, mostly through smaller phase 2 trials. A recent meta-analysis examined 49 studies and found that low-frequency stimulation to M1 on the unaffected hemisphere improved lower limb motor function, balance, and walking speed poststroke. The effects were stronger in the subacute than chronic phase poststroke.[8] Another meta-analysis suggested that rTMS improves upper extremity function as measured by the FMA-UE, particularly in the subacute phase poststroke, regardless of whether the contralesional or ipsilesional hemisphere was stimulated.[9] Not surprisingly, timing and severity of the stroke may impact response to rTMS, with those with severe strokes benefiting more.[9]

The impact of tDCS on motor impairment poststroke is slightly less clear. Although meta-analyses have indicated that tDCS can improve upper extremity motor outcomes,[10] particularly when coupled with constraint-induced movement therapy,[11] the recent high-quality RCT examining tDCS and constraint-induced movement therapy[12] calls this into question. Fewer studies have been conducted examining tDCS and gait, with a recent meta-analysis identifying very low-quality evidence from 7 studies that employed tDCS, in combination with task-specific training, to improve walking speed in stroke.

As with studies of other treatment approaches for stroke recovery, there is considerable variation in the response to these treatments among different recipients, for reasons that remain to be fully elucidated, but which are suspected to include corticospinal tract integrity poststroke, stroke lesion location and size as well as chronicity[8] of stroke and severity.[9] Variability in treatment response may also be impacted by the dose, timing and intensity, as well as paired rehabilitation therapy and concomitant drugs.[13] Better predictive tools for selecting individuals most likely to respond may improve the ability to demonstrate efficacy in the right subgroups. Moreover, the numerous variations in the different types of stimulation, targets in the brain and possibilities of combinations with other therapies would suggest that we are likely to continue to see many more trials in the years to come.

Robotics

A large number of RCTs have examined the impact of robotic devices on stroke recovery. Robotic rehabilitation devices vary substantially in their design. Two common classes of device are end-effector devices and exoskeleton devices. With end-effector devices, the patient

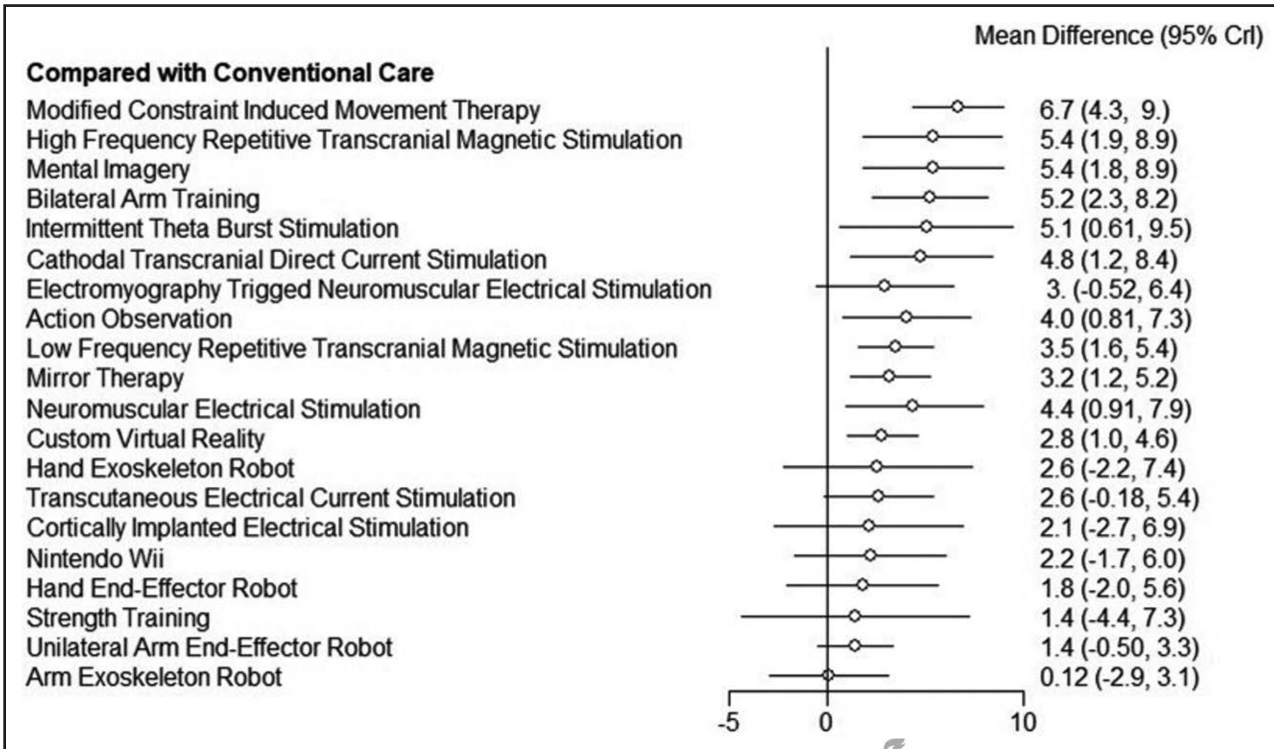

Figure 2. Network meta-analysis of the impact on Fugl-Meyer upper extremity assessment in randomized controlled trials of poststroke upper extremity interventions compared with conventional care.
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typically uses the most distal part of their limb (eg, hand/foot) to interact with a handle or single point of attachment on the end of the robotic device, whereas in an exoskeletal device, the patient wears the device and the device is positioned so that its joints align and move one-to-one with the user's joints (eg, wrapping around the arm or leg). Devices may focus on a particular joint or pair of joints (eg, shoulder and elbow) or the entire limb, with varying df. Thus, the device used, timing poststroke and the body part targeted by the treatment are just some of the potential factors that can influence study outcomes.

Most of the trials conducted in robotics have been relatively small, with a notable exception being the RATULS trial (Robot Assisted Training for the Upper Limb After Stroke[14]; N=770) in the upper extremity, which did not demonstrate a significant improvement in patients with the Massachusetts Institute of Technology (MIT)-Manus end-effector robot compared with usual care. Limitations of this study include modest doses (45 minutes, 3× per day, 12 weeks) of robotic treatment, lack of standardization of control treatment, and that most participants were enrolled in the chronic phase poststroke. Meta-analyses conducted before the RATULS trial (44 RCTs, n=1362), such as Veerbeek et al[15] and more recently Boardsworth et al[16] (54 RCTs, n=2744) and De Iaco et al[17] (90 RCTs, n=4311), have all suggested a small but significantly positive improvement in upper extremity impairment after robotic therapy for the upper limb (≈2 points on the FMA-UE), but concerns have been raised about how improvement in

Table 2. Number of Motor and Cognitive Randomized Controlled Trials Examining Noninvasive Brain Stimulation Up Until December 31, 2024

Brain Stimulation Interventions	UE	LE	GCD	Neglect	Aphasia	Total (excluding Duplicates)
rTMS total*	120	35	14	9	29	199
tDCS total	119	58	21	7	49	239
Theta burst stimulation†	30	8	6	10	6	59
Other brain stimulation RCTs	11	2	1	0	2	13
Total	280	101	42	26	85	510

GCD indicates general cognitive disorders; LE, lower extremity; RCT, randomized controlled trial; rTMS, repetitive transcranial magnetic stimulation; tDCS, transcranial direct current stimulation; and UE, upper extremity.

*1 RCT examines rTMS over cerebellum.

†Form of rTMS but considered as a separate intervention in our database.

Table 3. Number of Robotics Interventions Studied in Motor and Cognitive Randomized Controlled Trials

Intervention	UE	LE	Cognitive	Total
Exoskeletons	74	25	4	100*
Robot-assisted gait training	0	94	2	95*
End-effectors UE	91	0	4	93*
Electromechanical devices	15	10	1	25*
Robotic soft gloves	20	0	0	20
Actuated robots†	15	0	1	15*
Ankle robot training LE	0	13	1	13*
Robotic orthosis UE	10	0	0	10
Robot-assisted balance training	0	6	0	6
Combination of multiple devices	8	1	0	8*
Haptic devices	5	0	0	5
Robotic verticalization	0	2	1	2*
Total interventions	238	151	14	392*
Total RCTs	238	150	14	391*

One or more interventions can be examined in a single RCT. LE indicates lower extremity; RCT, randomized controlled trial; and UE, upper extremity.
*Excluding RCT duplicates across data sets.
†Includes singly actuated, wire-actuated, and pneumatically actuated.

impairment does not seem to lead to an improvement in activities of daily living (ADL).[17]

In the lower extremity, exoskeletons can further be classified into those that rely on a stationary platform such as a treadmill (eg, the Hocoma-Lokomat) versus exoskeletons designed to be worn for overground walking.[18] A meta-analysis combining both types of devices (23 RCTs, n=907) suggested pairing robot-assisted gait training with conventional rehabilitation improved gait speed, balance, and ADL performance and that end-effector robots produced superior outcomes to exoskeletons, particularly in subacute stroke. Overground robot-assisted gait training (ie, walking exoskeletons, treadmill weight-support devices) was shown in a meta-analysis (7 RCTs, n=288) to only slightly increase walking speed (0.009 m/s).[19] In a 2025 Cochrane review, Mehrholz et al[20] suggested that electromechanical-assisted gait training combined with conventional rehabilitation therapy is more likely to result in independent ambulation poststroke.

Generally, the promise of robotics is to deliver higher doses of activity practice without increasing the amount of care provided by healthcare personnel. These labor savings remain largely aspirational using currently available robotic systems but are becoming increasingly achievable as the technology advances. Providing larger doses of robotic therapy earlier poststroke will be needed to maximize the benefits and may be facilitated by devices intended for home use without real-time professional supervision. It is likely that we will see the field of robotics continue to evolve over the coming years with hardware and software advances (eg, artificial intelligence, brain-machine interfaces, lighter-weight wearable devices) driving the next generation of devices and clinical trials.

THERAPY INTENSITY AND DOSE

The concepts of therapy intensity and dose in rehabilitation are intertwined, are often context-dependent, and include time spent in therapy, [21] augmented therapy time,[22,23] number of repetitions, [24] heart rate, or perceived level of exertion. This is further complicated by variability in treatment protocols.[25] Increased therapy, however defined, and in different settings, improves motor deficits poststroke, including ADLs and walking speed.[22,23] Intensity of therapy may allow patients to achieve greater ADL independence faster through greater compensation of the nonparetic limb instead of neurological recovery.[26] Lohse et al[27] in a meta-analysis found increasing time spent in therapy strongly predicted overall functional improvement.

Therapy Intensity

Intensity of stroke rehabilitation therapy often refers to the frequency of repetitions of task-specific therapy conducted in active face-to-face therapy conducted with physical, occupational, or speech therapists, or it can refer to the time spent in therapy practicing task-specific therapy or even the level of effort during therapy sessions.

Repetitive Task-Specific Training

This involves performing repetitions of active motor sequences within a single training session, combining intensity and task specificity to improve specific functions.[28] Repetitive sit-to-stand exercises improved mobility[28] while increased number of repetitions during task-specific training of the upper extremity improved functional recovery.[29] Research suggests that hundreds of repetitions in task-specific practice may be required to optimize function poststroke.[29] The number of repetitions provided during actual poststroke rehabilitation is generally only a small fraction of the optimal number.[30]

High Intensity Exercise

High intensity interval training involves intermittent bursts of effort (eg, 30 seconds to 4 minutes) separated by periods of recovery,[31] and improves aerobic capacity[32] more quickly and efficiently. In a systematic review, Wiener et al[31] found that high intensity interval training, using a treadmill or stationary bike, resulted in significant improvements in walking speed and endurance as well as balance. Adverse effects were rare and minor. High intensity interval training on treadmills led to significant improvements in walking speed and motor evoked potentials (electrical signals recorded by stimulating the motor cortex), while high intensity resistance training improves paretic leg strength relative to lower intensity training.[33]

Therapy Dosing

Dose of therapy refers to the total amount of therapy activity and can refer to the duration of a single rehabilitation therapy session but more often refers to the total amount of therapy delivered. Dosing has often been equated with intensity, in particular the number of hours or additional hours spent in therapy.[34] It is a combination of the intensity of therapy and frequency of therapy sessions. Kwakkel et al[22] performed a meta-analysis and found that during the first 6 months poststroke, a 16-hour increase in therapy time over usual care was associated with a favorable outcome. Van Peppen et al[35] noted that an additional therapy time of 17 hours over 10 weeks is necessary for significant positive effects; this was affirmed by Veerbeek et al[36] Similarly, Dromerick et al[37] in the Critical Period After Stroke Study found 20 extra hours of self-selected task-specific motor therapy when provided in the subacute and to a lesser degree the acute phase, significantly improved Action Research Arm Test, but not in the chronic phase.

The DOSE trial (Determining Optimal Post-Stroke Exercise) was designed to study the effect of higher exercise doses on walking for rehabilitation patients 1 to 4 weeks poststroke.[38] The study consisted of 3 groups—a control group receiving standard physiotherapy for 1 h/d, 5 d/wk for 4 weeks, DOSE1 receiving 1 hour of more intensive physiotherapy during the same period, and DOSE2 who received 2 hours of more intensive physiotherapy per day, 5 d/wk for 4 weeks; this amounts to an additional 20 to 40 hours of more intensive therapy respectively replacing the previous standard therapy. Both DOSE groups showed greater walking endurance at 4 weeks and at 1-year follow-up compared with controls.

Clark et al[39] in a Cochrane review, found the difference in total time spent in therapy between control and intervention groups ranged from 186 to 6160 minutes with a median difference of 840 minutes (14 hours). The authors concluded there was insufficient high-quality evidence to determine the effect of time spent in rehabilitation to recommend a minimum beneficial daily amount in clinical practice, with the lack of studies with a large contrast in additional therapy (minimum of 1000 minutes or 16.7 hours) between intervention and control groups being a limitation.

The ideal amount of therapy has never been well-defined, and guidelines from different countries differ regarding therapy time recommendations. Generally, the amount of therapy can be defined as the number of hours of exercises/tasks provided with direct therapist engagement; recent evidence strongly suggests that an additional 20 hours or more of therapy overall improves functional outcomes, particularly in selected patients.

In the United States, 3 hours of therapy per day is often stated as the standard and is associated with better outcomes when compared with <3 hours.[40,41] However, achieving 3 hours of therapy on an inpatient rehabilitation unit is not always feasible, given many patients are older or have medical comorbidities. MacDonald et al[40] conducted a retrospective real-world study of 12 770 individuals admitted to inpatient rehabilitation facilities in Ontario, Canada from January 2017 to December 2021. Increased rehabilitation intensity (total therapy min/d) was associated with better outcomes for total, motor and cognitive Functional Independence Measure scores, rehabilitation effectiveness and 90-day home time. However, for individuals receiving more than 95 minutes of rehabilitation therapy per day, there was markedly diminishing returns for all outcomes except the Functional Independence Measure cognitive subscale, which continued to increase with additional time in rehabilitation therapy without diminishing returns. This suggests a need to better identify which individuals are most likely to benefit from higher doses of therapy.

Challenges in Achieving Increased Intensity

Many clinicians are prevented from providing an ideal amount of rehabilitation therapy because of lack of resources. Canadian stroke rehabilitation guidelines recommended 180 minutes of physiotherapy, occupational therapy, and speech-language therapy. In contrast, the median amount of therapy in inpatient rehabilitation in Ontario increased from 57 min/d in 2015/2016 to 69 min/d in 2019/2020, which was well below recommended targets. In stroke rehabilitation inpatient units serving ≥ 40 patients in 2019/2020, therapy varied from 22 to 110 min/d, with no unit meeting the guidelines.[42] Hence, within a single payer health care system, there was extreme variability in the amount of therapy based on location of the inpatient unit. Similar variability has been observed in the United States, where in one study, the percentage of acute stroke patients discharged to an inpatient rehabilitation facility varied from 0% to 71%.[43] On average, it has been reported that one quarter of acute United States stroke patients go to an inpatient stroke rehabilitation facility and one quarter go to a skilled-nursing facility.[44] Of those who go to inpatient stroke rehabilitation facilities, rehabilitation stays are limited to an average of 2 weeks. Of those who are discharged directly home, access to therapy is very limited.[44] This is indicative of a resource crisis in stroke rehabilitation which runs contrary to evidence demonstrating the positive impact of stroke rehabilitation therapy intensity, particularly in the acute/subacute phase.

The rehabilitation outcomes utilized tend to be focused on activities, such as ADLs, and not impairment or body structure function outcome measures. That means they are susceptible to decline in the absence of continued use and practice. An excellent example is constraint-induced movement therapy motor improvements seen when the individual is forced to perform

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tasks with the paretic arm; these improvements were often not well generalized into the home environment. An exception was when there was a transfer package.[45] This highlights the importance of linking the stroke survivors' preferences and values to the goals of practice and having a means to maintain gains made with more intensive therapy.

Stroke rehabilitation hospital units can be demotivating. De Wit et al[46] observed that patients spent 72% of their time engaged in nontherapeutic activities on average. Further, an Australian study by Simpson et al[47] found that patients spent more time upright and walking during the first week at home compared with their last week of rehabilitation. This suggests that stroke patients may be discouraged from achieving their full activity potential while on organized stroke rehabilitation units, possibly related to safety concerns.

TIMING OF REHABILITATION INTERVENTIONS POSTSTROKE

Timing of therapy refers to time poststroke onset and is traditionally divided into acute, subacute, and chronic phases. Timing of therapy impacts patient outcomes, likely through greater neuroplasticity early on.

Most Stroke Rehabilitation RCTs Are Conducted in the Chronic Phase

When looking at the stroke rehabilitation research, one is immediately struck by the fact that in 3602 motor and cognitive RCTs, over 50% are conducted in the chronic phase (Figure S1), defined as >6 months poststroke.[48] The challenge with conducting stroke rehabilitation in the chronic phase is that it clashes with animal studies, which have shown the damaged brain shows maximal response to therapy when initiated early after the stroke.[49] However, the precise timing of this early window of increased neuroplasticity has not been fully determined.[50] A recent network meta-analysis[51] examining changes in FMA-UE score showed interventions studied in the chronic phase (≥6months poststroke) had less than half of the impact of the same interventions when studied in RCTs conducted in the acute/subacute phase (<6months poststroke), a result influenced by lesser neuroplasticity in chronic phase RCTs, such that these studies run the risk of significantly underestimating the impact of the same intervention conducted in the acute/subacute phase.[51] However, those same studies did demonstrate that improvement still takes place when therapy is applied in the chronic phase. Most rehabilitation clinically takes place in the acute/subacute phase, and the results of RCTs conducted in the chronic phase are much less generalizable to actual clinical care. This tendency to conduct clinical trials in the chronic phase is most prominent in North America (73.1% of RCTs), and anecdotally, when researchers are questioned about this, they report that it is easier to conduct studies later, once participants are no longer engaging in active rehabilitation and recruitment seems easier outside of institutional constraints. Clinically, research conducted in the chronic phase provides support for treatment of residual deficits, particularly with new adjunct therapies or even a combination of therapies, much like cross-training in sports medicine. However, when dealing with chronic stroke, in many cases, there have already been significant changes in the brain/neural networks, and it requires up to 3.5× as much therapy.[25]

TIMING AND DOSE IN POSTSTROKE REHABILITATION

Although intensity and dose in stroke rehabilitation matter, the interplay of timing and dose is also very important.

Increased Dose in the Subacute Phase

Dromerick et al[37] in the Critical Period After Stroke Study specifically examined the optimal time for motor recovery. Twenty extra hours of self-selected task-specific motor therapy were provided to 3 different groups each at different time intervals poststroke: acute (<30 days); subacute (2–3months); and chronic (>6 months) with a standard rehabilitation control group. On the Action Research Arm Test, a test of arm function, the greatest improvement was seen in the subacute phase; there was a significant but smaller improvement in the acute phase and a nonsignificant improvement in the chronic phase poststroke. This study established that additional therapy was best provided in the subacute phase, more than 30 days poststroke, and is consistent with some of the dose studies described earlier. One potential concern is that early treatment focusing on compensatory strategies using the nonparetic limb may reduce opportunities for neurological restoration and paretic limb recovery while prioritizing functional independence.[52]

Increased Dose in the Chronic Phase

Over 50% of motor-cognitive RCTs are conducted in the chronic phase of stroke. Lohse et al[27] in a systematic review, found that in 34 RCTs studying 1750 chronic stroke patients, those receiving more therapy did better. On average, the control groups received only 40% of the therapy received by the intervention group. Increased therapy in the chronic phase can improve rehabilitation outcomes, but there appears to be a higher threshold. The ICARE trial (Interdisciplinary Comprehensive Arm Rehabilitation Evaluation), a large multisite RCT, compared outpatients who received 30 hours of a structured task-specific upper extremity exercise program over 10 weeks and found no significant difference in outcomes when compared with those receiving usual care.[53] Similarly, Lang et al[29] found 32 hours of therapy in the chronic phase of stroke did not improve upper extremity function. In contrast, in a single blind interventional study looking at upper extremity recovery, Daly et al[54] found that chronic stroke patients who were provided 300 hours of activity-based technology-assisted therapy (5 h/d, 5 d/wk, for 12 weeks) showed a substantial improvement in the FMA-UE score above the minimum clinically important difference. Similarly, in an observational trial by Ward et al,[55] 224 patients at a median of 18 months poststroke were treated with a high-intensity rehabilitation program, 90 hours of therapy delivered for 6 h/d, 5 d/wk, for 3 weeks, and compared with 2 lower-intensity therapy groups. The high intensity rehabilitation program resulted in significant improvement in several arm motor outcome measures, including the Action Research Arm Test and the FMA-UE.

On reviewing the role of timing of more intensive therapy poststroke, a pattern of different thresholds emerges. Augmented therapy of about 20 hours in total is enough to improve motor outcomes in the acute/subacute phase. In the chronic phase of stroke, more hours of therapy are required, up to 90 to 300 additional hours, to result in similar motor recovery outcomes. This is consistent with our meta-analysis findings that timing of RCTs matters because neuroplasticity matters and baselines are higher, with RCTs being conducted in the chronic phase undervaluing the impact of the same intervention studied in the acute/subacute phase.[51]

PERSONALIZED CARE

Stroke patients vary in their impairments and disabilities; size, location of stroke, and response to rehabilitation are highly variable and influenced by psychosocial factors. This heterogeneity means there are some limits to standardization of care and an increasing role for individualized care. The most common variable currently utilized is clinical measures of stroke severity, in particular the degree of functional disability, which forms the basis for benchmarking, where length of inpatient rehabilitation care is often dictated by algorithms based on anticipated response to therapy based on these severity scores. Despite the simplicity and popularity of this model, it has limitations.

Alternatively, predicting outcomes has been left to experienced clinicians, which then plays a role in determining the treatment approach. However, Rinske et al,[56] as part of the EPOS trial (Early Prediction of Functional Outcome After Stroke), found experienced physiotherapists were only 60% accurate in predicting recovery of arm function when assessments were made within 72 hours poststroke. This has led to the use of technology in the form of noninvasive biomarkers which can be superior to behavioral examinations in predicting stroke recovery and outcome.[57] Increasingly, research has been promoting more physiologically precise mechanisms, such as transcranial magnetic stimulation, to examine the intactness of the corticospinal tract,[58] which when combined with clinical measures can be used to predict upper extremity recovery.[59] Being able to better predict response to rehabilitation would allow more effective use of more intensive therapy resources for identified patients.

GLOBAL CHANGES IN STROKE REHABILITATION CLINICAL RESEARCH

Stroke rehabilitation clinical research is changing globally, which has implications for how research evidence is generated and applied. Global trends indicate that the number of RCTs from LMICs has been rising, compared with high-income countries (HICs).[60] Individuals in HICs are more likely to have access to early rehabilitation and mobilization, compared with those living in LMICs.[61] In addition, there are large variations in the proportion of stroke survivors who receive rehabilitation, as well as the number of in-hospital rehabilitation sessions in LMICs, leading to higher rates of severe functional disability, compared with stroke survivors living in HICs.[28]

Up to the end of 2024, 69.5% of motor and cognitive RCTs published in English-language journals came from HICs and 30.5% arose from LMICs; each year LMICs account for a higher percentage of the overall research. Figure 3 shows a plateauing of annual publication rates of RCTs in HICs since 2014 after a decade of growth. This prolonged plateau remains unexplained. We speculate that possible reasons include reaching maximal capacity in HICs, increasingly arduous and complicated regulations and ethics requirements, increased costs of conducting trials in HICs, reduced research funding or diversion of funding to other priorities, and fewer researchers willing or able to conduct clinical trials for the above or alternative reasons, such as a greater emphasis on social programming. Other reasons may be related to regulations aimed at improving transparency by mandating preregistration, as well as data dissemination and sharing.[62]

RCTs from LMICs have similar quality scores, in this case assessed by using the Physiotherapy Evidence Database score,[63] and modestly greater sample sizes than RCTs from HICs but tend to be published in lower-impact journals.[1] LMICs also tend to publish more in the acute/subacute phase.[1] For instance, motor-cognitive RCTs conducted in the chronic phase account for 13.3% in Mainland China, compared with 74.2% in the United States and 68.2% in Canada.The impressive rise of RCTs from LMICs and the unexplained plateauing from HICs has important implications, as LMIC research is

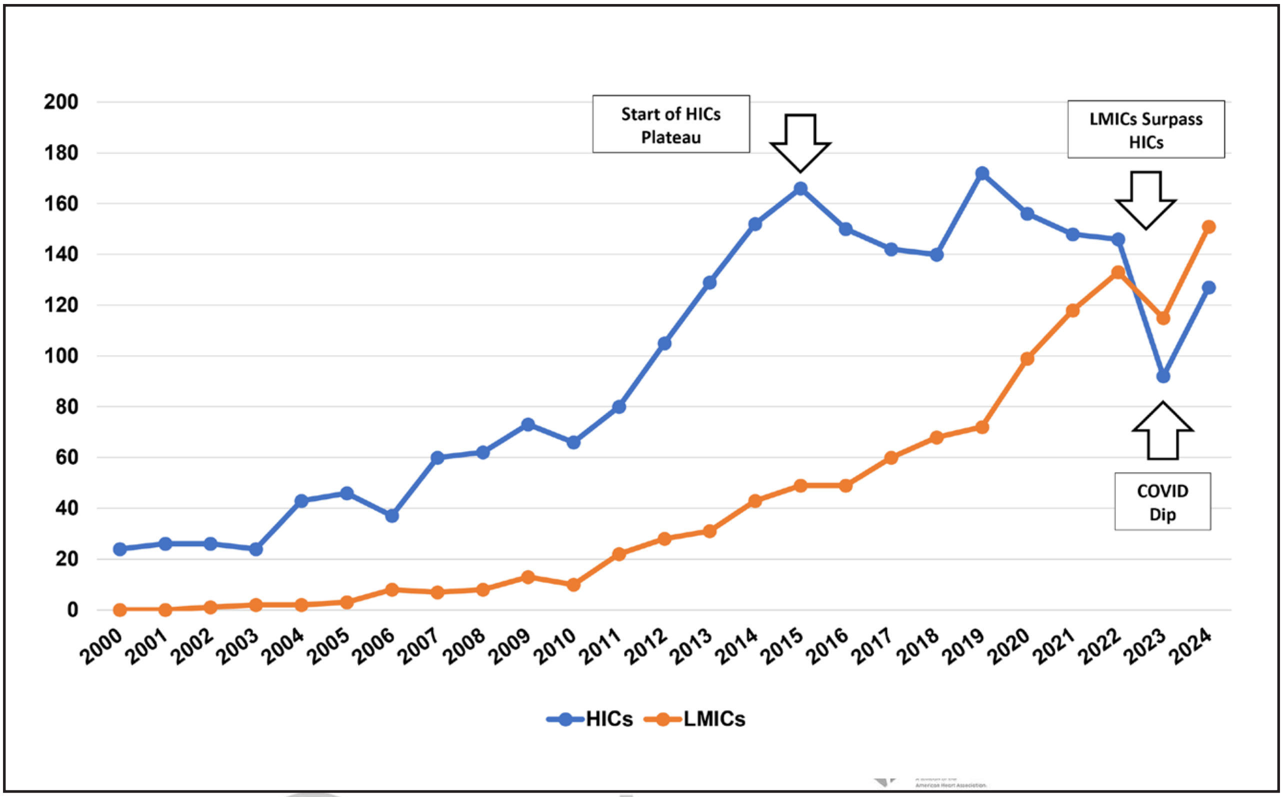

Figure 3. Annual number of motor-cognitive randomized controlled trials from high-income countries (HICs) and low- and middle-income countries (LMICs), from 2000 to 2024.

conducted in a lower resource environment, and if current trends continue, LMICs will be the primary source of future research studies in stroke rehabilitation. Although it has been argued that findings from LMIC RCTs may not be directly translatable to HICs, a much higher percentage of RCTs from LMICs are conducted in the acute/subacute phase where therapy actually takes place in clinical practice.

CONCLUSIONS

The stroke rehabilitation research landscape has been evolving rapidly, with just over 3600 motor and cognitive RCTs to the end of 2024, making it the most prolifically studied area of neurorehabilitation. This is to some extent offset by a large number of smaller underpowered RCTs, of which half are conducted in the chronic phase. Network meta-analyses are possible now for the more common outcome measures, offering a form of comparative analysis of the efficacy of different interventions for a single outcome measure. Therapy intensity and dosage are regarded as important to improving recovery, but the optimal dose is still uncertain. However, timing matters when it comes to therapy intensity and dosage; therapy conducted in the chronic phase, when neuroplasticity has declined, requires a much higher dose to have the same impact as the same therapy delivered in the acute/subacute phase. Personalized care, made more feasible by new technologies, may help to better identify which patients are more likely to benefit from more intensive therapy. Globally, the annual number of RCTs from LMICs has been exceeding those from HICs; this is important because LMICs used different outcome measures and conducted more trials in the acute/subacute phase. The rapid rise of RCTs from LMICs is something to celebrate; the decade-long plateau in RCTs from HICs is of concern and raises concerns about research infrastructure and processes.

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Disclosures

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Supplemental Material

Table S1
Figure S1

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